Measurement Conversions

A big unit holds many small units. Multiply to change units.

1 meter is 100 centimeters

$3\ \text{m} = 3 \times 100 = 300\ \text{cm}$

Big unit to small unit: multiply.

Make a table

ft	in
1	12
2	24
3	36
4	48
5	60

1 foot = 12 inches.
Each row adds 12 more.
5 ft = 5 × 12 = 60 in.

Units to know

1 km = 1,000 m 1 m = 100 cm 1 kg = 1,000 g 1 L = 1,000 mL
1 yd = 3 ft 1 ft = 12 in 1 lb = 16 oz
1 hr = 60 min 1 min = 60 sec 1 day = 24 hr 1 week = 7 days

Mixed units

$2\ \text{m}\ 35\ \text{cm} = 200\ \text{cm} + 35\ \text{cm} = 235\ \text{cm}$

Your turn!

$4\ \text{ft} = \boxed{48}\ \text{in}$

For teachers

Goal: change a bigger unit into a smaller one and solve measurement word problems (4.MD.1–2). Say "How many small units in one big unit? Then how many in 3?" Make a table.

Measurement Conversions: Practice

Big unit to small unit: multiply by how many fit in one.

Part 1 Fill in the table The first row is done.

1)

m	cm
1	100
2	200
4	400
6	600
9	900

2)

yd	ft
1	3
3	9
5	15
7	21
10	30

3)

lb	oz
1	16
2	32
3	48
5	80
8	128

4)

hr	min
1	60
2	120
4	240
5	300
10	600

Part 2 Metric units No help this time.

5) 4 m = 400 cm

6) 3 kg = 3,000 g

7) 5 L = 5,000 mL

8) 2 km = 2,000 m

9) 7 m = 700 cm

10) 9 kg = 9,000 g

Measurement Conversions: Practice (continued)

Part 2 Metric units (continued)

11) 6 km = 6,000 m

12) 8 L = 8,000 mL

13) 12 m = 1,200 cm

14) 4 kg = 4,000 g

15) 3 L = 3,000 mL

16) 10 km = 10,000 m

Part 3 Feet, pounds and time No help this time.

17) 3 ft = 36 in

18) 4 yd = 12 ft

19) 2 lb = 32 oz

20) 5 min = 300 sec

21) 3 hr = 180 min

22) 2 yd = 72 in

Measurement Conversions: Practice (continued)

Part 3 Feet, pounds and time (continued)

23) 6 ft = 72 in

24) 3 weeks = 21 days

25) 4 lb = 64 oz

26) 2 days = 48 hr

27) 8 yd = 24 ft

28) 7 min = 420 sec

Part 4 Mixed units No help this time. Change the big unit, then add.

29) 3 m 42 cm = 342 cm

30) 1 kg 250 g = 1,250 g

31) 3 ft 4 in = 40 in

32) 2 hr 15 min = 135 min

33) 4 m 8 cm = 408 cm

34) 2 L 600 mL = 2,600 mL

Measurement Conversions: Practice (continued)

Part 4 Mixed units (continued)

35) 5 ft 6 in = 66 in

36) 1 hr 45 min = 105 min

37) 2 yd 1 ft = 7 ft

38) 3 lb 5 oz = 53 oz

Part 5 Which is more? No help this time. Change to the same unit first.

39) 3 ft < 40 in

40) 2 m > 150 cm

41) 1 kg < 1,200 g

42) 5 yd > 14 ft

43) 2 hr > 100 min

44) 3 L = 3,000 mL

45) 2 lb < 40 oz

46) 4 min > 200 sec

Measurement Conversions: Practice (continued)

Part 6 Add and take away No help this time. Use the small unit.

47) 1 L − 350 mL = 650 mL

48) 2 m + 45 cm = 245 cm

49) 3 ft − 8 in = 28 in

50) 1 hr + 25 min = 85 min

51) 1 kg − 375 g = 625 g

52) 2 yd + 2 ft = 8 ft

53) 1 km − 400 m = 600 m

54) 1 lb + 9 oz = 25 oz

Part 7 Convert

55) 4 min = 240 sec

56) 8 km = 8,000 m

57) 10 hr = 600 min

Measurement Conversions: Practice (continued)

Part 7 Convert (continued)

58) 3 lb = 48 oz

59) 8 min = 480 sec

60) 12 hr = 720 min

61) 7 L = 7,000 mL

62) 8 yd = 288 in

63) 3 km = 3,000 m

64) 8 lb = 128 oz

65) 4 weeks = 28 days

66) 11 m = 1,100 cm

67) 10 yd = 30 ft

68) 11 L = 11,000 mL

69) 5 yd = 180 in

Measurement Conversions: Practice (continued)

Part 7 Convert (continued)

70) 11 weeks = 77 days

71) 12 ft = 144 in

72) 3 m = 300 cm

73) 9 lb = 144 oz

74) 3 min = 180 sec

75) 9 min = 540 sec

76) 2 hr = 120 min

77) 9 ft = 108 in

78) 11 min = 660 sec

79) 3 yd = 9 ft

80) 9 yd = 324 in

81) 11 kg = 11,000 g

Measurement Conversions: Practice (continued)

Part 7 Convert (continued)

82) 8 kg = 8,000 g

83) 11 hr = 660 min

84) 4 hr = 240 min

85) 8 m = 800 cm

86) 11 days = 264 hr

87) 5 yd = 15 ft

88) 11 yd = 396 in

89) 9 hr = 540 min

90) 3 yd = 108 in

Part 8 Remember? Problems from earlier packets.

91) 0.7 > 0.65

92) $\frac{38}{100}$ = 0.38

Measurement Conversions: Practice (continued)

Part 8 Remember? (continued)

93) $5 \times \frac{3}{4} = \frac{15}{4}$

94) $2\frac{5}{6} + 1\frac{3}{6} = \frac{4}{2}$ $\frac{}{6}$

95) $36 \times 25 =$ 900

96) $4{,}568 \div 8 =$ 571

Part 9 Word problems Change to the same unit first.

97) A rope is 3 m long. Leo cuts off 75 cm. How long is the rope now, in centimeters?

Number sentence 300 cm − 75 cm = 225 cm

Answer 225 cm

Draw or show your work

98) A movie is 2 hours long. Ana has watched 85 minutes. How many minutes are left?

Number sentence 120 − 85 = 35

Answer 35 minutes

Draw or show your work

99) A bag of apples weighs 2 kg. Each apple is about 200 g. About how many apples are in the bag?

Number sentence 2,000 ÷ 200 = 10

Answer 10 apples

Draw or show your work

100) Mia drinks 250 mL of milk each day. How much does she drink in a week?

Number sentence 7 × 250 = 1,750

Answer 1,750 mL

Draw or show your work

Measurement Conversions: Practice (continued)

Part 9 Word problems (continued)

101) A desk is 4 ft long. A shelf is 50 in long. Which is longer?

Number sentence 4 ft = 48 in < 50 in

Answer shelf

Draw or show your work

102) Challenge: Ben runs 400 m around the track 6 times. Is that more than 2 km?

Number sentence 6 × 400 = 2,400 m > 2,000 m

Answer yes

Draw or show your work

Check-In: Measurement Conversions

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Fill in the table

1)

min	sec
1	60
2	120
3	180
6	360
8	480

Part 2 Change the unit

2) 7 yd = 252 in

3) 5 lb = 80 oz

4) 2 kg = 2,000 g

5) 5 hr = 300 min

6) 2 m = 200 cm

7) 12 days = 288 hr

Part 3 Mixed units

8) 4 m 5 cm = 405 cm

9) 6 ft 1 in = 73 in